



**COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF ICT**

DEPARTMENT OF INFORMATION SYSTEMS

LEVEL 4

PROJECT TITLE:

**Design and Implementation of a Hospital-to-Hospital patient data exchange platform
(Case study for CHUK and King Faisal Hospital)**

Submitted by group 18

Names:

- | | |
|---------------------------------|------------------|
| 1. IZERE Mpuhwe Bede | 222011744 |
| 2. ISHIMWE KWIZERA Irene | 222010573 |
| 3. IZABAYO Claudien | 222003541 |

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Abstract

Background

Rwanda's national referral hospitals, including CHUK and King Faisal Hospital, use computerized hospital information systems and relational databases to store patient data. Despite this digital capacity, there is no mechanism for direct electronic exchange of patient information between these hospitals. During referrals, critical data is often transferred using paper-based documents or manually generated reports, leading to fragmented information, delayed care, repeated tests, and potential medical errors. This reveals a gap between digital data storage and effective digital data exchange within the healthcare system.

Methods

The project began with an assessment of existing patient data management systems, database structures, and referral workflows at CHUK and King Faisal Hospital to identify interoperability challenges. Based on the findings, a centralized system architecture was designed to function as an interoperability layer connecting the independent hospital databases without replacing them. The platform enables standardized exchange of key patient information, including demographics, medical history, laboratory results, imaging reports, and treatment details.

Results

The implemented platform successfully enabled secure, real-time sharing of patient data between CHUK and King Faisal Hospital during referrals. Authorized clinicians were able to access relevant patient records from the originating hospital's database through the centralized exchange layer. The system reduced reliance on paper-based data transfer, and enhanced continuity of care by providing timely access to comprehensive patient information.

Conclusion

The project demonstrates that a centralized interoperability platform can effectively bridge independent hospital databases and support secure inter-hospital patient data exchange without replacing existing systems. This solution improves clinical decision-making, and continuity of care. The platform provides a scalable model that can be extended to other healthcare institutions in Rwanda, contributing to the country's broader digital health transformation.

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CHAPTER I: GENERAL INTRODUCTION

1.1 Background

The healthcare system in Rwanda is organized in a structured referral hierarchy that includes health posts, health centers, district hospitals, provincial hospitals, and national referral hospitals. At the highest level of this hierarchy are national referral hospitals such as Centre Hospitalier Universitaire de Kigali (CHUK) and King Faisal Hospital, which provide specialized and tertiary healthcare services to patients referred from across the country.

Healthcare governance and policy formulation are managed by the Ministry of Health, while implementation and technical oversight are largely handled by institutions such as the Rwanda Biomedical Centre (RBC). In recent years, Rwanda has demonstrated strong commitment to digital health transformation through the adoption of electronic medical records (EMRs), hospital information systems (HIS), and national health information platforms.

At the hospital level, institutions such as CHUK and King Faisal Hospital use computerized hospital information systems supported by relational database management systems (RDBMS) to store and manage patient data. These systems typically rely on centralized relational databases (such as SQL Server and Oracle) to store structured patient information including demographics, diagnoses, laboratory results, prescriptions, and clinical notes. Each hospital maintains its own database instance hosted within its internal infrastructure or secured data centers.

While these database systems are effective for internal hospital operations, they are designed to function independently. The databases at CHUK and King Faisal Hospital are not directly connected, synchronized, or interoperable. As a result, patient data is stored in separate silos, with each hospital maintaining its own version of patient records using different database schemas, data models, and system configurations.

In the current referral workflow, when patients are transferred between hospitals (particularly between CHUK and King Faisal), patient information is commonly exchanged through paper-based referral letters, printed laboratory results, CDs for imaging data, or manually generated

reports. Even where digital records exist, access to these records is restricted to the originating hospital's database system. This limits the ability of receiving hospitals to electronically retrieve complete and up-to-date patient information.

Given that CHUK and King Faisal Hospital frequently manage complex cases that require continuity of care across facilities, the lack of a shared or interoperable database mechanism presents significant operational challenges. Clinicians at receiving hospitals often begin treatment without access to prior medical history stored in the originating hospital's database, despite that data existing electronically.

This situation highlights a critical gap between digital data availability and digital data exchange within Rwanda's healthcare system. Although hospitals use modern database systems to store patient information, there is no centralized or federated data exchange platform that enables secure, standardized, and real-time sharing of patient data between hospitals.

Therefore, the design and implementation of a centralized hospital-to-hospital patient data exchange digital platform that will be capable of interfacing with existing relational databases at CHUK and King Faisal Hospital is essential. Such platform would act as an interoperability layer, allowing hospitals (CHUK and King Faisal) to securely exchange patient data without replacing their existing database systems, by improving referral efficiency, continuity of care, and overall healthcare service delivery in Rwanda.

1.2 Problem statement

Despite Rwanda's significant progress in adopting digital health systems, the healthcare sector continues to face challenges related to interoperability and data sharing between hospitals.

National referral hospitals such as CHUK and King Faisal Hospital use computerized hospital information systems supported by relational database management systems to store patient data electronically. However, these database systems operate independently and are designed primarily to support internal hospital operations.

Currently, there is no centralized or standardized digital mechanism that enables CHUK and King Faisal Hospital to securely exchange patient data directly from their existing databases. As

a result, patient information remains fragmented across separate hospital systems, even when patients are referred between these two major referral hospitals.

During inter-hospital referrals, patient data is often transferred through paper-based referral forms, printed reports, or manually generated summaries, despite the fact that complete medical records already exist in electronic databases at the originating hospital. This disconnect between digital data storage and digital data exchange leads to several challenges, including delayed referrals, incomplete patient information at receiving hospitals, and repeated diagnostic tests.

Furthermore, clinicians at receiving hospitals frequently lack timely access to critical patient information such as medical history, laboratory results, imaging reports, allergies, and ongoing treatments. This can compromise clinical decision-making, particularly in emergency cases, and increases the risk of medical errors.

Although CHUK and King Faisal Hospital utilize modern database systems capable of storing structured patient data, the lack of an integration layer or central exchange platform prevents these systems from communicating with one another. This highlights a critical gap in Rwanda's healthcare digital infrastructure: the absence of a secure hospital-to-hospital patient data exchange platform that can integrate existing databases without replacing them.

1.3 Project objectives

1.3.1 General Objective

To design and implement a centralized digital platform that enables secure, real-time hospital-to-hospital patient data exchange between CHUK and King Faisal Hospital.

1.3.2 Specific Objectives

The specific objectives of this project are:

- ✓ To assess and document the existing patient data management systems, database structures, and referral workflows used at CHUK and King Faisal Hospital.
- ✓ To identify interoperability gaps and challenges that prevent effective electronic exchange of patient data between the two hospitals.

- ✓ To design a centralized system architecture that enables secure integration and communication between the independent relational databases used by CHUK and King Faisal Hospital.
- ✓ To develop a hospital-to-hospital patient data exchange platform that allows authorized clinicians to access shared patient records during referrals.
- ✓ To implement standardized data exchange mechanisms that support consistent and structured sharing of patient information between the two hospitals.

1.3.3 Project Questions and Hypothesis

The research and development of this platform are driven by the need to address the following key questions and their corresponding hypothesis:

Q1. System Assessment: What are the current architectures, database structures, and referral workflows at CHUK and King Faisal Hospital, and how do they currently manage patient data during transfers?

H1: Currently, both CHUK and King Faisal Hospital (KFH) utilize independent relational database management systems (like SQL Server or Oracle) for internal operations. It is hypothesized that patient data management during transfers relies heavily on manual, paper-based processes such as physical referral letters and printed reports, rather than direct digital synchronization.

Q2. Interoperability Challenges: What specific technical and operational gaps exist that prevent these independent hospital information systems from communicating with each other?

H2: The primary gaps are the lack of a shared or federated data exchange platform and the use of different database schemas and data models at each institution. These technical silos, combined with restricted access to records outside the originating hospital's infrastructure, prevent real-time electronic communication.

Q3. Architectural Design: How can a centralized system architecture be designed to serve as an interoperability layer that connects independent relational databases (SQL Server, Oracle) without requiring their replacement?

H3: A centralized interoperability layer can be designed to interface with existing databases through secure APIs or data interfaces. This architecture will allow for data retrieval and sharing by acting as a bridge between the hospitals without altering their internal, independent system configurations.

Q4. Functional Development: What specific technical mechanisms and interfaces are required to allow authorized clinicians to securely submit referrals and access shared patient records in real-time?

H4: Functional development requires a web-based clinician interface, secure authentication and access control mechanisms, and a central exchange API. These tools will enable authorized staff to submit digital referral requests and view real-time data retrieved directly from the originating hospital's database.

Q5. Standardization: What data exchange standards are necessary to ensure that information such as demographics, laboratory results and imaging are shared in a consistent and structured format?

H5: Standardized data exchange mechanisms and logical mapping of patient data fields are necessary to ensure consistency. By adopting standards similar to global benchmarks like HL7 or IHE profiles, the platform can ensure that clinical data (medical history, lab results, and imaging) is structured uniformly for both institutions.

1.4 Scope of the project

This project focuses on the design and implementation of a centralized hospital-to-hospital patient data exchange digital platform with a clearly defined institutional, functional, technical, and operational scope.

1.4.1 Institutionally

the study will be conducted at Centre Hospitalier Universitaire de Kigali (CHUK) and King Faisal Hospital, these two hospitals are selected as national referral hospitals that frequently exchange patient referrals, The platform will be implemented as a pilot system intended to demonstrate feasibility of inter-hospital data exchange.

1.4.2 Functionally

the system will support the electronic exchange of selected patient data required during referrals, including: Patient demographic information, medical history and diagnoses, Laboratory test results, Imaging and investigation reports, Medication and treatment information and the platform will enable: Secure submission of referral requests, Authorized access to shared patient records by clinicians, Viewing of patient data during inter-hospital transfers.

1.4.3 Technically

the platform will be designed to integrate with existing hospital database systems at CHUK and King Faisal Hospital, Existing hospital information systems and databases will not be replaced, A centralized exchange layer will be implemented to facilitate data sharing between the two hospitals.

Primary users will include: Clinicians involved in patient referrals, Authorized hospital staff managing referrals

1.5 Project schedule (Gantt chat)

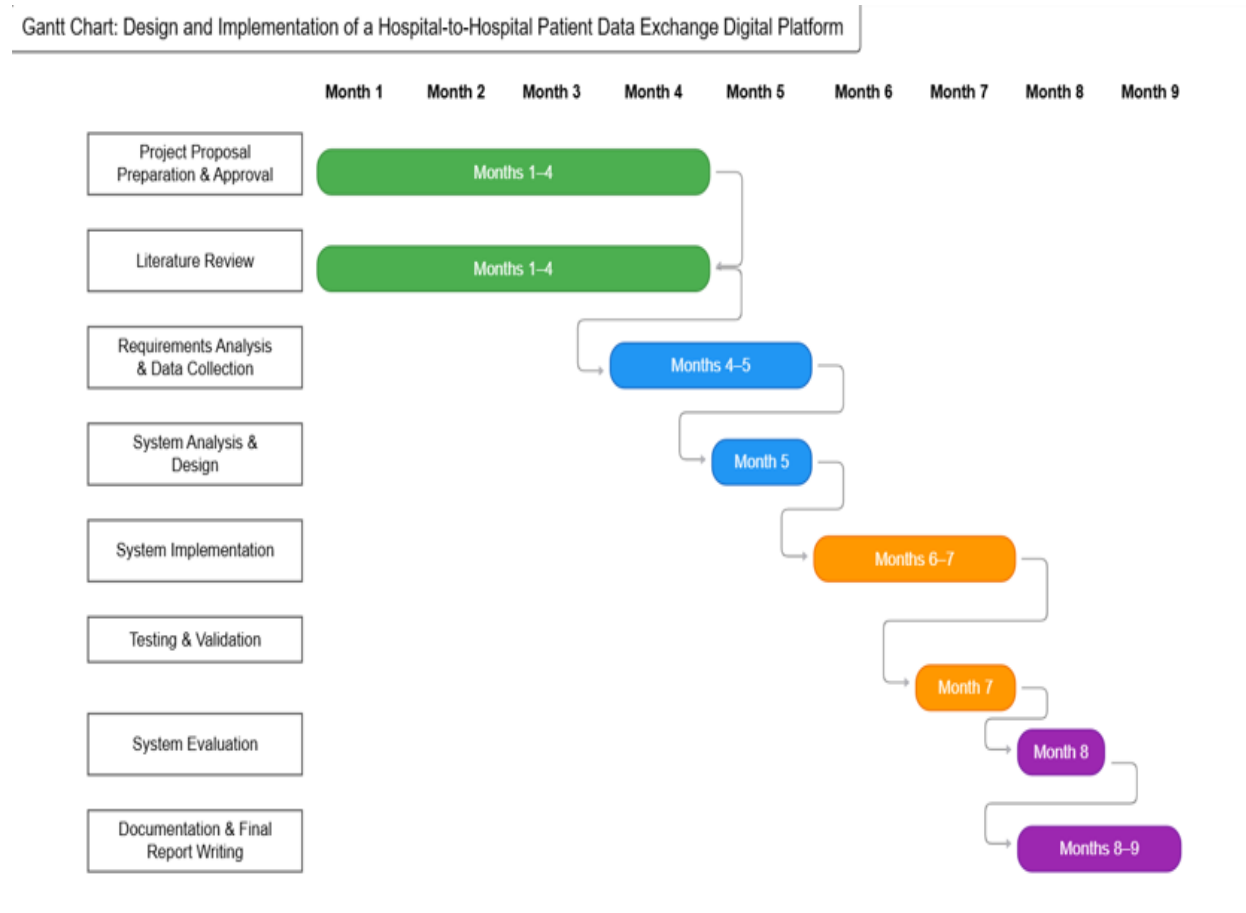


Figure 1.5 Project schedule or Gantt chart

1.6 Organization of the project

The is divided and organized in three chapters:

- ✓ **CHAPTER I:** deals with the background of project, problem statement, Research objectives, Scope of project and project organization.
- ✓ **CHAPTER II:** deals with literature review introduction, existing similar works/systems, gaps identified and expected project outcomes.
- ✓ **CHAPTER III:** deals with methodology will be used, introduction, data collection techniques, software development methodology and system analysis and design method.

CHAPTER II: LITERATURE REVIEW

2.1 Introduction

The literature review explores the conceptual and practical foundations of health information exchange (HIE) systems. It examines global standards for interoperability and evaluates existing models of data sharing to provide a benchmark for the proposed digital platform connecting CHUK and King Faisal Hospital.

2.2 Existing similar works/systems

The eHealth Exchange is one of the largest and most successful health information exchange networks in the United States. Originally a federal program under the Office of the National Coordinator for Health IT (ONC), it is now an independent, non-profit network that connects federal agencies and private sector healthcare providers.

Feature	Description
Model	A network-of-networks that uses a standardized approach to facilitate the secure exchange of health information across the country.
Participants	Includes the Department of Veterans Affairs (VA), the Department of Defense (DoD), the Social Security Administration, and thousands of private hospitals and pharmacies.
Standardization	Relies on a Common Agreement and a set of performance specifications, ensuring all participants use the same technical standards (such as HL7 and IHE profiles).
Functionality	Supports query-based exchange (finding a patient's record at another facility) and "push" notifications for transitions of care.

2.3 Gaps Identified

While systems like the eHealth Exchange provide a gold standard for national interoperability, significant gaps remain within the Rwandan (CHUK and KFH) context:

- **Regional Silos:** Current systems at CHUK and King Faisal Hospital are designed primarily for internal operations and rely on independent relational databases.
- **Paper Dependency:** Unlike the eHealth Exchange's automated queries, Rwandan referrals still rely heavily on paper-based documents and manually generated reports.
- **Lack of Interoperability Layer:** There is currently no centralized or federated data exchange platform in Rwanda that can interface with existing SQL Server or Oracle databases without replacing them.
- **Fragmented Information:** Clinical data (labs, imaging, history) exists digitally but is inaccessible to clinicians at the receiving hospital during the critical referral window.

2.4 Expected Project Outcomes

By referencing the standardized communication models of the eHealth Exchange, this project aims to achieve the following:

- ✓ **Centralized Interoperability Layer:** Implementation of a system architecture that functions as a bridge between independent hospital databases.
- ✓ **Real-Time Data Access:** Authorized clinicians will be able to retrieve comprehensive patient records, including demographics: lab results, and imaging, in real-time.
- ✓ **Reduced Clinical Risk:** Enhanced continuity of care and improved decision-making by eliminating fragmented information and repeated diagnostic tests.

CHAPTER III: METHODOLOGY

3.1 Introduction

This chapter presents the methodology that will be used in the design and implementation of the hospital-to-hospital patient data exchange digital platform for CHUK and King Faisal Hospital. It describes the research approach, data collection techniques, software development methodology, and system analysis and design methods to be applied to achieve the project objectives.

3.2 Data Collection Techniques

To ensure that the system meets real hospital needs, both primary and secondary data collection techniques to be used.

3.2.1 Primary Data Collection

Primary data will be collected directly from stakeholders involved in patient referrals at CHUK and King Faisal Hospital. The following techniques will be used:

- **Interviews:** Semi-structured interviews will be conducted with hospital staff and clinicians to understand existing referral workflows, patient data handling procedures, and challenges related to inter-hospital data exchange.
- **Observation:** Direct observation of referral processes will be carried out to identify how patient information is currently transferred between hospitals.

3.2.2 Secondary Data Collection

Secondary data will be obtained from existing documents and online resources, including:

- Hospital reports and referral forms.
- Published research articles on health information systems and interoperability.
- Ministry of Health and Rwanda Biomedical Centre (RBC) digital health policy documents.

3.3 Software Development Methodology

The Waterfall software development methodology will be adopted for this project. The Waterfall model is a structured and sequential approach to system development, where each phase is completed before moving to the next.

The Waterfall methodology will be adopted and tailored to suit the academic nature of the project. The methodology will focus on requirements analysis, system design, implementation, testing, evaluation, and Documentation. Deployment and long-term maintenance phases are excluded since the system is developed as a prototype intended for academic demonstration and feasibility analysis rather than full-scale production use.

The Waterfall methodology will be implemented through the following phases:

- **Requirements Analysis:** In this phase, system requirements will be collected from clinicians and hospital staff at CHUK and King Faisal Hospital. The focus will be on understanding referral workflows, patient data requirements, and interoperability challenges.
- **System Design:** Based on the collected requirements, the overall system architecture, database integration mechanisms, security controls, and user interface designs will be developed.
- **Implementation:** This phase involved the actual development of the hospital-to-hospital patient data exchange platform, including creation of the centralized exchange layer and secure interfaces with existing hospital databases.
- **Testing and Validation:** After implementation, the system will be tested to ensure correct functionality, data accuracy validation, security validation, and reliability. Testing will focus on validating referral processes and patient data exchange between CHUK and King Faisal Hospital.
- **Evaluation:** In this phase, the developed system will be evaluated against the project objectives to assess its effectiveness in supporting secure and efficient inter-hospital patient data exchange.

- **Documentation:** In this final phase, System documentation will include technical descriptions and user guidelines, will also be prepared to support understanding and future enhancement of the platform.

3.4 System Analysis and Design Methods

To ensure a structured and reliable system, standard system analysis and design methods will be applied.

3.4.1 System Analysis

System analysis focus on understanding the existing hospital information systems at CHUK and King Faisal Hospital, identifying interoperability gaps, and defining functional and non-functional requirements of the proposed platform.

Key activities included:

- Analysis of existing database structures
- Identification of data elements required for referrals
- Definition of user roles and access levels

3.4.2 System Design

System design involves translating requirements into a technical solution. The following design components will be developed:

- **System Architecture Design:** A centralized exchange layer that connects independent hospital databases without replacing them.
- **Database Design:** Logical mapping of patient data fields to enable standardized data exchange.
- **Security Design:** Implementation of authentication, authorization, and access control mechanisms to ensure data confidentiality.
- **User Interface Design:** Simple and intuitive interfaces for clinicians to submit referrals and access patient records.

3.5 System Diagrams

To clearly illustrate how the proposed hospital-to-hospital patient data exchange platform operates, several system diagrams were developed. These diagrams help in understanding the interaction between users, the system, and existing hospital databases.

3.5.1 System Architecture Diagram

The system architecture is based on a centralized interoperability layer that connects the independent hospital information systems of CHUK and King Faisal Hospital. Existing hospital databases are not replaced; instead, they are securely interfaced with the exchange platform.

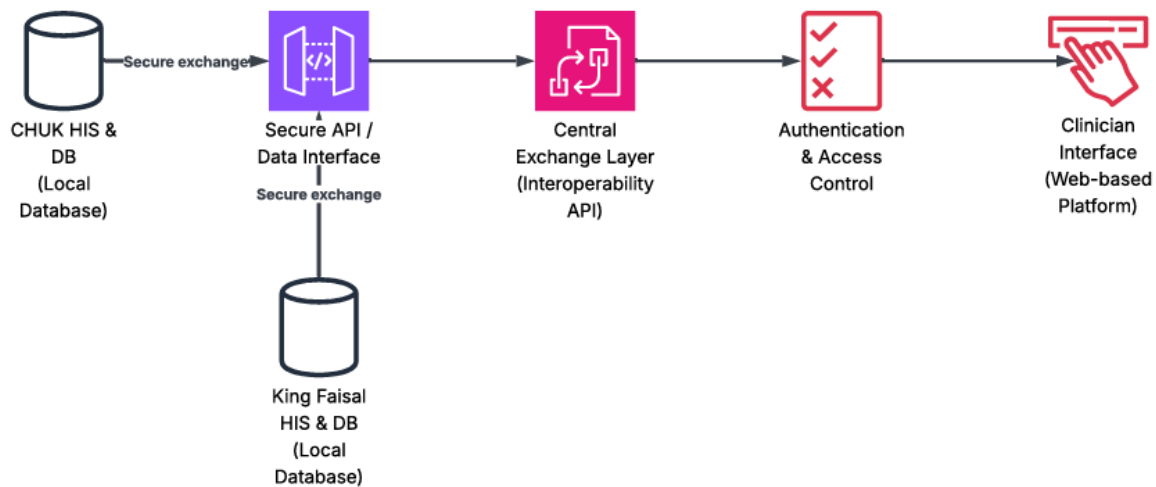


Figure 3.5.1 System Architecture Diagram

3.5.2 Use Case Diagram

The use case diagram shows interactions between system users and the patient data exchange platform.

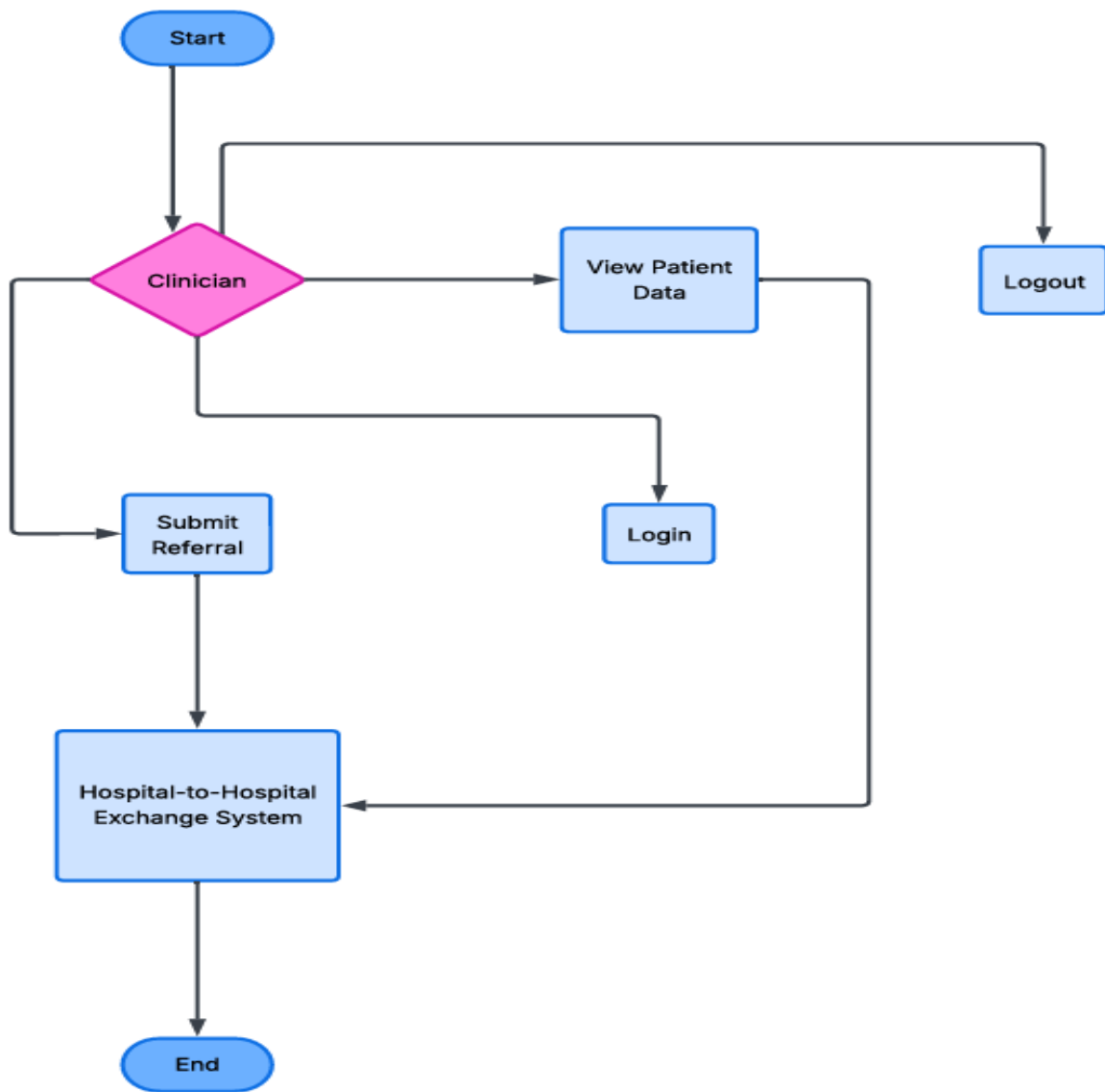


Figure 2.5.2 Use Case Diagram

3.5.3 Data Flow Diagram (Level 0)

The Data Flow Diagram illustrates how patient data moves between hospitals through the centralized platform.

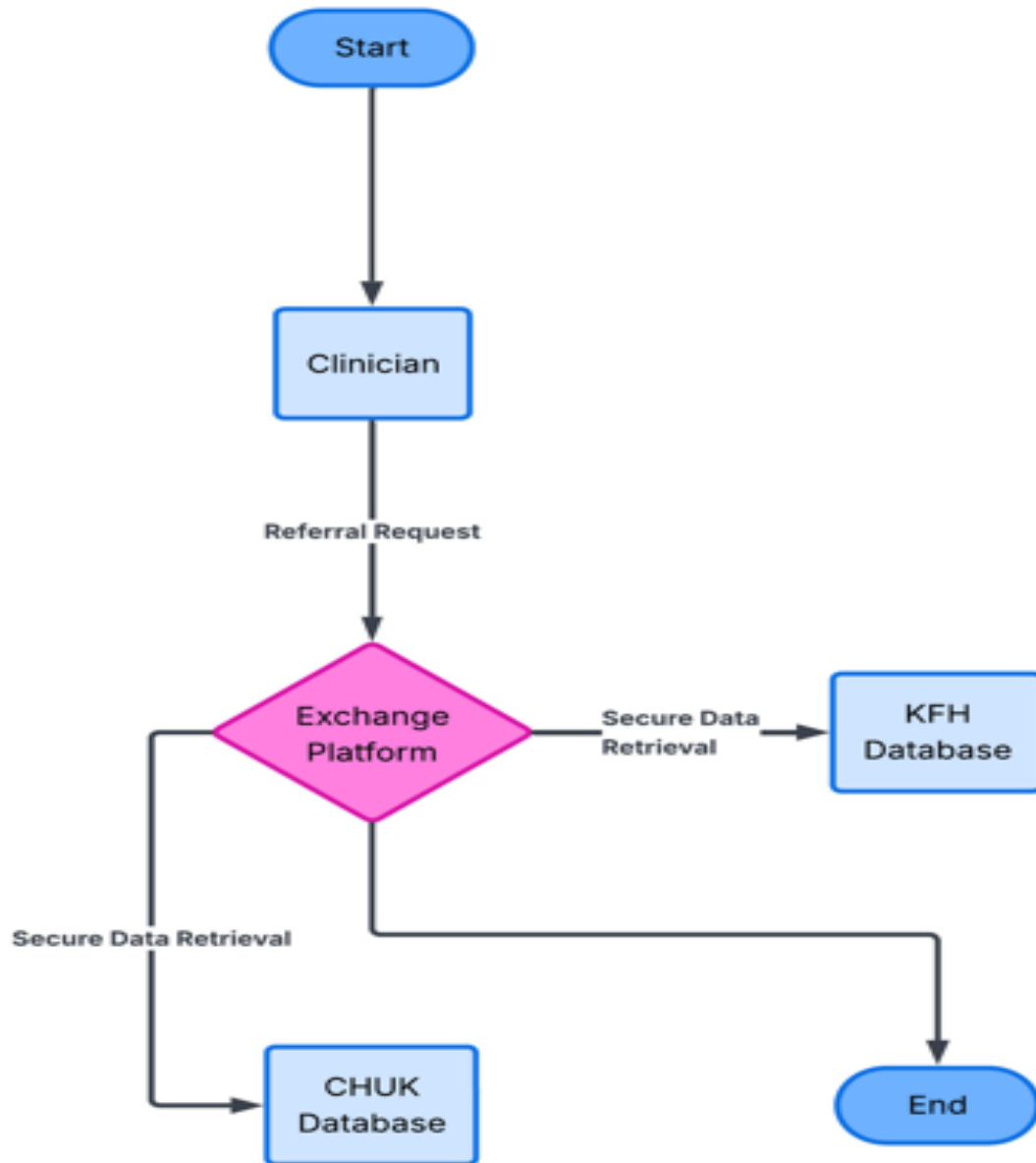


Figure 3.5.3 Data Flow Diagram (Level 0)

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